

Problem Solving Kinematics

One-dimensional kinematics involves only five variables—position (or displacement, Δx), initial velocity (v_0), final velocity (v), acceleration (a), and time (t)—linked by a small set of equations. The challenge students face is rarely the physics itself; it is deciding which equation to use and keeping track of what is actually known. A systematic six-step method removes that guesswork.

Step 1 — Identify the governing principle. Confirm the motion is one-dimensional with constant acceleration, which justifies using the standard kinematic equations rather than calculus-based variable-acceleration methods.

Step 2 — List the knowns. Extract every numerical value given, assigning consistent signs based on a chosen positive direction.

Step 3 — Identify the unknown. State exactly what the problem asks for.

Step 4 — Select the equation that contains the unknown and only known quantities—avoiding unnecessary multi-step chains.

Step 5 — Substitute values with units and solve algebraically before plugging in numbers, then compute.

Step 6 — Verify reasonableness, checking units, sign, and order of magnitude.

Worked example. A car accelerates from rest at 2.5 m/s^2 for 6 s. Find the distance traveled.

- *Knowns:* $v_0 = 0$, $a = 2.5 \text{ m/s}^2$, $t = 6 \text{ s}$
- *Unknown:* Δx
- *Equation:* $\Delta x = v_0 t + \frac{1}{2} a t^2$
- *Substitute:* $\Delta x = 0 + \frac{1}{2} (2.5) (6)^2 = 45 \text{ m}$
- *Check:* 45 m in 6 s from rest under modest acceleration is physically sensible—neither absurdly small nor unreasonably large.

Problem-solving application. The real payoff of this method appears in multi-part or multi-object problems, such as two cars starting at different times or a ball

thrown upward that later falls back down. By re-running the six steps for each phase of motion—resetting knowns and unknowns at each transition (e.g., velocity at the top of a throw is zero, but time continues accumulating)—students avoid the most common error: reusing a variable's value from one phase in the equation for another. Treating each stage as a fresh six-step problem, connected only by shared boundary conditions (final velocity of phase one becomes initial velocity of phase two), turns seemingly complex kinematics problems into a sequence of manageable, verifiable calculations.